Computer Organization and Architecture (CSL251)
LAB-5
2025-26 W

Instructions:

1. File Naming

Q1_ReverseNumber_XX.s,
Q2_PalindromeNumber_XX.s, etc. (XX = roll number)

2. Submission Format

Zip all .s files in a single folder:

RISC-V_Assignment/
- Q1_ReverseNumber_XX.s
- Q2_PalindromeNumber_XX.s
- Q3_SumOfDigits_XX.s
- Q4_CountDigits_XX.s

3. Screenshot

Take a screenshot showing register output after running each program in Spike.

Name screenshots to match the program:

Q1_ReverseNumber_XX_output.png
Q2_PalindromeNumber_XX_output.png

Your final folder should look like this:

RISC-V_Assignment/
- Q1_ReverseNumber_XX.s
- Q1_ReverseNumber_XX_output.png
- Q2_PalindromeNumber_XX.s

Q2_PalindromeNumber_XX_output.png
Q3_SumOfDigits_XX.s
Q3_SumOfDigits_XX_output.png
Q4_CountDigits_XX.s
Q4_CountDigits_XX_output.png

Question 1: Reverse a Number

Write a RISC-V assembly program to reverse a given integer.
The number is hard-coded in the program.

Input: Number stored in register t0

Output : Reversed number stored in register t1

Example

Input: 1234

Output: 4321

Question 2: Palindrome Number

Write a RISC-V assembly program to check whether a number is a palindrome.

Input Number stored in register t0

Output t6 = 1 → Palindrome

t6 = 0 → Not Palindrome

Example

Input: 121 → Palindrome

Input: 123 → Not Palindrome

Question 3: Sum of Digits

Write a RISC-V assembly program to find the sum of digits of a given number.

Input Number stored in register t0

Output Sum of digits stored in register t1

Example

Input: 456

Output: 15

Question 4: Count Digits in a Number

Write a RISC-V assembly program to count the number of digits in a given integer.

Input: Number stored in register t0

Output: Digit count stored in register t1

Example

Input: 98765

Output: 5

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